Mendel Gödel Machine: Recursive Self-Improving Coding Agents via Comparative Evolution

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HarnessRSI 65 ★★

evidence: retained

Paper

Code

1 What changes; what is kept

1 Archive + inherited code
Source, ancestry, trajectories, task outcomes.
Model weights stay fixed.

2 Sample evaluation or expansion
Node outcomes guide evaluation; descendant outcomes guide lineage expansion (HGM).

3 Choose eligible evidence
CM: one failure. RM: same agent, two tasks.
CH: different agents, same task.

4 Agent edits its own scaffold
CH adapts a reference behavior, not files.
If reference succeeds, edit the failing agent.

5 Admit valid child → archive
Validity gate; no immediate parent-beating score threshold is specified (Algorithm 1).

6 Evaluate → update beliefs → select
Binary outcomes guide future search.
Report best-belief agent, not raw maximum.

Finite search: 200 evaluations + 24 expansions. Valid code persists; performance affects sampling and final choice.

Example lineage: read tests/stubs → extract API contract → implement → bounded test/repair loop (up to 5 attempts).

Scope: scaffold evolves; no demonstrated autonomous rewrite of the outer search policy. Successive edits compose, but long-horizon compounding/saturation is not established.

2 Largest gain: Polyglot

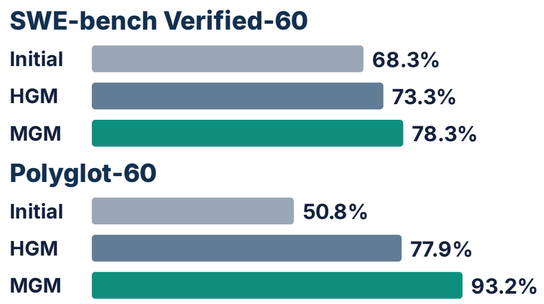


Table 1 · Qwen3.6-35B-A3B · best-belief agents. Same ancestor and operation counts for HGM/MGM.

MGM over HGM: **15.3‡ points** on Polyglot, **5.0 points** on SWE-bench. Evolution-subset scores, not independent held-out tasks.

4 Where evidence is thinner

- **Coder-Next-80B (Table 7):** initial 33.3%, HGM 40.0%, MGM 41.7%; small margin, no uncertainty estimate.
- **Not wholly held out:** frozen full Polyglot scores 210/225 (93.3%), but includes evolution tasks. Private tests are denied; available repo tests are read/run. No contamination audit.
- **Run scope:** empirical tables give no per-cell evolution-run count, SD or CI. Benchmark/model averages are not seed means.
- **Unresolved:** F.2 prose says 61.7% for Qwen MGM; Tables 1/7 say 78.3%. Other variants do not resolve it; we use tables.

3 Controls, transfer and cost

Ablation · Polyglot-60 (Table 4)

200 evaluations	Accuracy	Hours
Full MGM	93.2%	40.14
Without RM	79.7%	43.01
Without CH	74.6%	39.95

Same ancestor, two workers; no seed SD/CI. HGM already has archive + adaptive sampling, but CM only. No failed-pool-only control.

Frozen transfer · initial / HGM / MGM

New benchmark (Table 2)	Accuracy %
SWE-bench Pro	16.7 / 13.3 / **26.7**
SWE-bench Multilingual	41.7 / 43.3 / **55.0**

60 tasks each, ChatGPT-selected with language coverage. No further self-improvement.

New model (Table 3)	Accuracy %
DeepSeek-V4-Flash	50.0 / 60.0 / **66.7**
DeepSeek-V4-Pro	45.0 / 70.0 / **75.0**

Frozen SWE scaffold; same Verified-60 tasks, backbone replaced. Qwen uses thinking mode; DeepSeek non-thinking (Appendix C.1). Inference budgets are not demonstrated matched across models; within-target-model comparisons remain.

Costs and unreported choices

HGM / MGM hours, eight H100s: SWE **93.02 / 96.11**; Polyglot **44.20 / 40.14**. Counts matched, not exact token totals; no total dollar cost.

Default-selection split and baseline tuning budgets unreported. Failed-pool weight **1.0** means no relative boost (our reading of Eq. 17).

Simulation only: 100 seeds / 95% CIs. At no fix-quality advantage, MGM becomes HGM-like; oracle is a diagnostic bound, not a coding competitor.

Takeaway

Comparing archived attempts improves inherited code edits at matched search counts. Frozen transfer is the strongest generality check; gains are larger on Polyglot and smaller with Coder-Next. This is retained scaffold improvement, not evidence of runaway recursive growth.

Our read · HarnessRSI

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Reports: matched counts, operator ablations, frozen transfer, evolution time.
Not reported: empirical seed uncertainty, tuning split/budgets, contamination audit.